

T3. Asteroid (10 points)

A peculiar asteroid of mass, m , was spotted at a distance, d , from a star with mass, M . The magnitude of the asteroid's velocity at the time of the observation was $v = \sqrt{\frac{GM}{d}}$, where G is the universal gravitational constant. The distance d is much larger than the radius of the star.

For both of the following items, express your answers in terms of M , d , and physical or mathematical constants.

(a) (8 points) If the asteroid is initially moving exactly towards the star, how long will it take for it to collide with the star?

(b) (2 points) If the asteroid is instead initially moving exactly away from the star, how long will it now take for it to collide with the star?

T4. White Dwarf (10 points)

The structure of a white dwarf is sustained against gravitational collapse by the pressure of degenerate electrons, a phenomenon explained by quantum physics and related to the Pauli Exclusion Principle for electrons. The equation of state of a gas made of non-relativistic degenerate electrons is the following:

$$P = \left(\frac{3}{8\pi}\right)^{2/3} \frac{h^2}{5m_e} n_e^{5/3},$$

where n_e is the number of electrons per unit volume, which can be expressed in terms of the mass density ρ using the dimensionless factor μ_e , the number of nucleons (protons and neutrons) per unit electron. Also consider that the central pressure can be described by this equation of state.

In the condition of hydrostatic equilibrium, the pressure and gravitational forces balance each other at any distance r from the centre of the star. This condition can be expressed by:

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2},$$

where $M(r)$ is the mass contained in the sphere of radius r , and $\rho(r)$ is the mass density of the star at a radius r .